Exercises ISLR – Ch.10

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20/10/2025

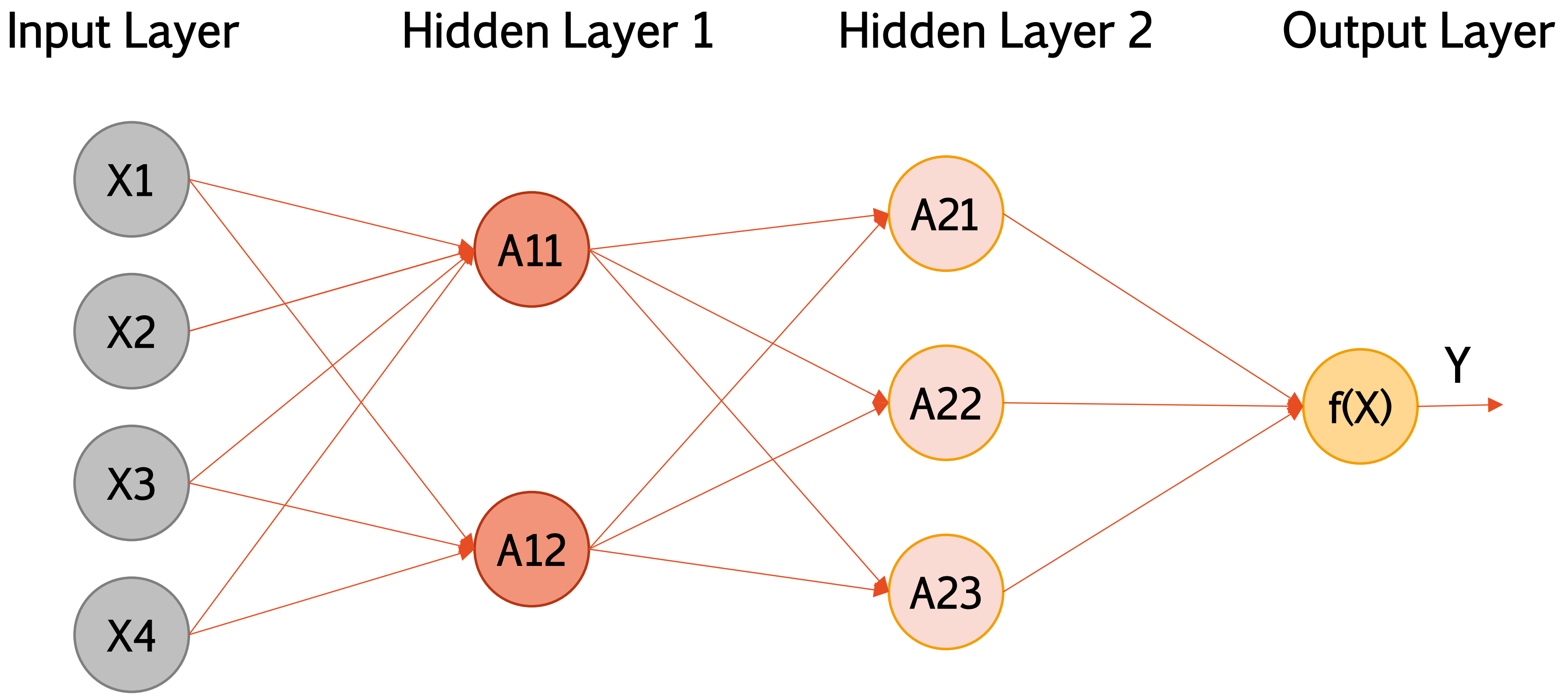
Conceptual

Exercises: 1, 4

Ex.1) Neural Network: Picture (a)

1. Consider a neural network with two hidden layers: $p = 4$ input units, 2 units in the first hidden layer, 3 units in the second hidden layer, and a single output.

   (a) Draw a picture of the network, similar to Figures 10.1 or 10.4.



Ex.1) Neural Network: Output (b)

(b) Write out an expression for $f(X)$, assuming ReLU activation functions. Be as explicit as you can!

Input Vector $X = (X_1, X_2, X_3, X_4)^T$

Hidden Layer 1 $f_j^1 = w_{0j} + w_{1j}X_1 + w_{2j}X_2 + w_{3j}X_3 + w_{4j}X_4, \quad j = 1{,}2$

$$f_j^1 = \sum_{i=0}^{4} w_{ij}X_i \quad j = 1{,}2$$

Linear Model: 4 predictors + 1 intercept (per cell) → **10 weights**

$$A_j^1 = max\left(0, \sum_{i=0}^{4} w_{ij}X_i\right) \quad j = 1{,}2$$

ReLU activation

Ex.1) Neural Network: Output (b)

(b) Write out an expression for $f(X)$, assuming ReLU activation functions. Be as explicit as you can!

Hidden Layer 2

$$f_j^2 = v_{0j} + v_{1j}A_1^1 + v_{2j}A_2^1, \quad j = 1,2,3$$

$$f_j^2 = \sum_{i=0}^{2} v_{ij}A_i^1 \quad j = 1,2,3$$

Linear Model: 2 predictors + 1 intercept (per cell) → **9 weights**

$$A_j^2 = max\left(0, \sum_{i=0} v_{ij}A_i^1\right) \quad j = 1,2,3$$

ReLU activation

Output Layer

$$Y = z_0 + z_1A_1^2 + z_2A_2^2 + z_3A_3^2$$

$$Z = \sum_{i=0}^{3} z_iA_i^2$$

Linear Model: 3 predictors + 1 intercept (per cell) → **4 weights**

Ex.1) Neural Network: Example (c)

(c) Now plug in some values for the coefficients and write out the value of $f(X)$.

Input Vector $X = (X_1, X_2, X_3, X_4)^T = \boxed{(\mathbf{1}, \mathbf{1}, \mathbf{0}, \mathbf{0})^T}$

Hidden Layer 1

$$\boxed{\begin{aligned} W_1 &= [\mathbf{0} \quad \mathbf{1} \quad \mathbf{1} \quad \mathbf{1} \quad \mathbf{1}] \\ W_2 &= [\mathbf{0} \quad -\mathbf{1} \quad -\mathbf{1} \quad -\mathbf{1} \quad -\mathbf{1}] \end{aligned}}$$

$$A_1^1 = max(0, 0 + 1 + 1 + 0 + 0) = \mathbf{2}$$

$$A_2^1 = max(0, 0 - 1 - 1 - 0 - 0) = \mathbf{0}$$

Ex.1) Neural Network: Example (c)

(c) Now plug in some values for the coefficients and write out the value of $f(X)$.

Hidden Layer 2

$A^1 = (\mathbf{2}, \mathbf{0})^T$

$$\boxed{\begin{aligned} V_1 &= [\mathbf{0} \quad \mathbf{1} \quad \mathbf{1}] \\ V_2 &= [\mathbf{0} \quad -\mathbf{1} \quad -\mathbf{1}] \\ V_3 &= [\mathbf{0} \quad \mathbf{1} \quad \mathbf{0}] \end{aligned}}$$

$$A_1^2 = max(0, 0 + 2 + 0) = \mathbf{2}$$

$$A_2^2 = max(0, 0 - 2 - 0) = \mathbf{0}$$

$$A_2^3 = max(0, 0 + 2 + 0) = \mathbf{2}$$

Output Layer

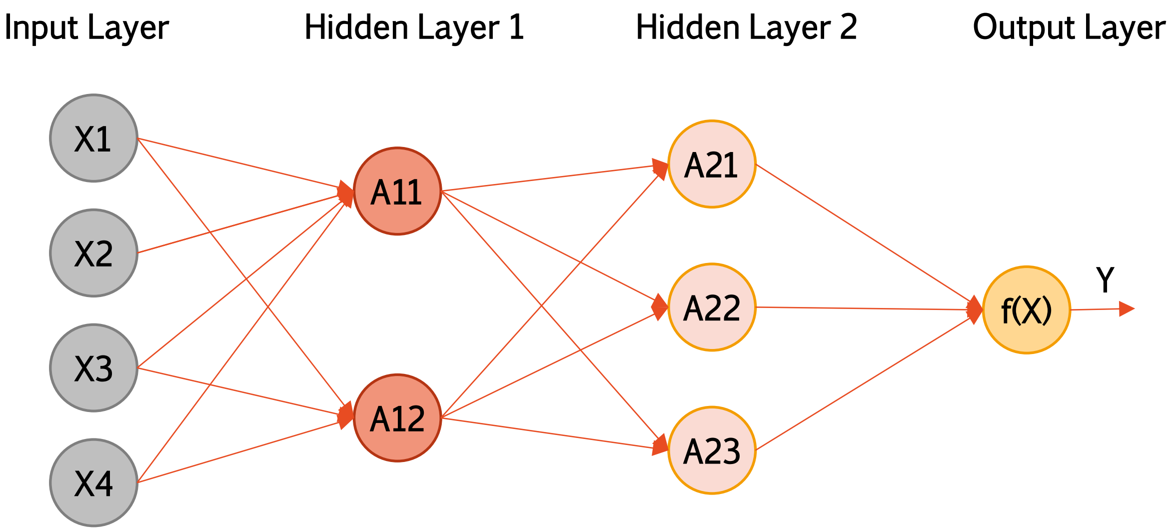
$A^2 = (\mathbf{2}, \mathbf{0}, \mathbf{2})^T$

$$\boxed{Z = [\mathbf{1} \quad \mathbf{1} \quad \mathbf{1} \quad \mathbf{1}]}$$

$$\mathrm{Y} = 1 + 2 + 0 + 2 \boxed{= \mathbf{5}}$$

Ex.1) Neural Network: Output (d)

(d) How many parameters are there?

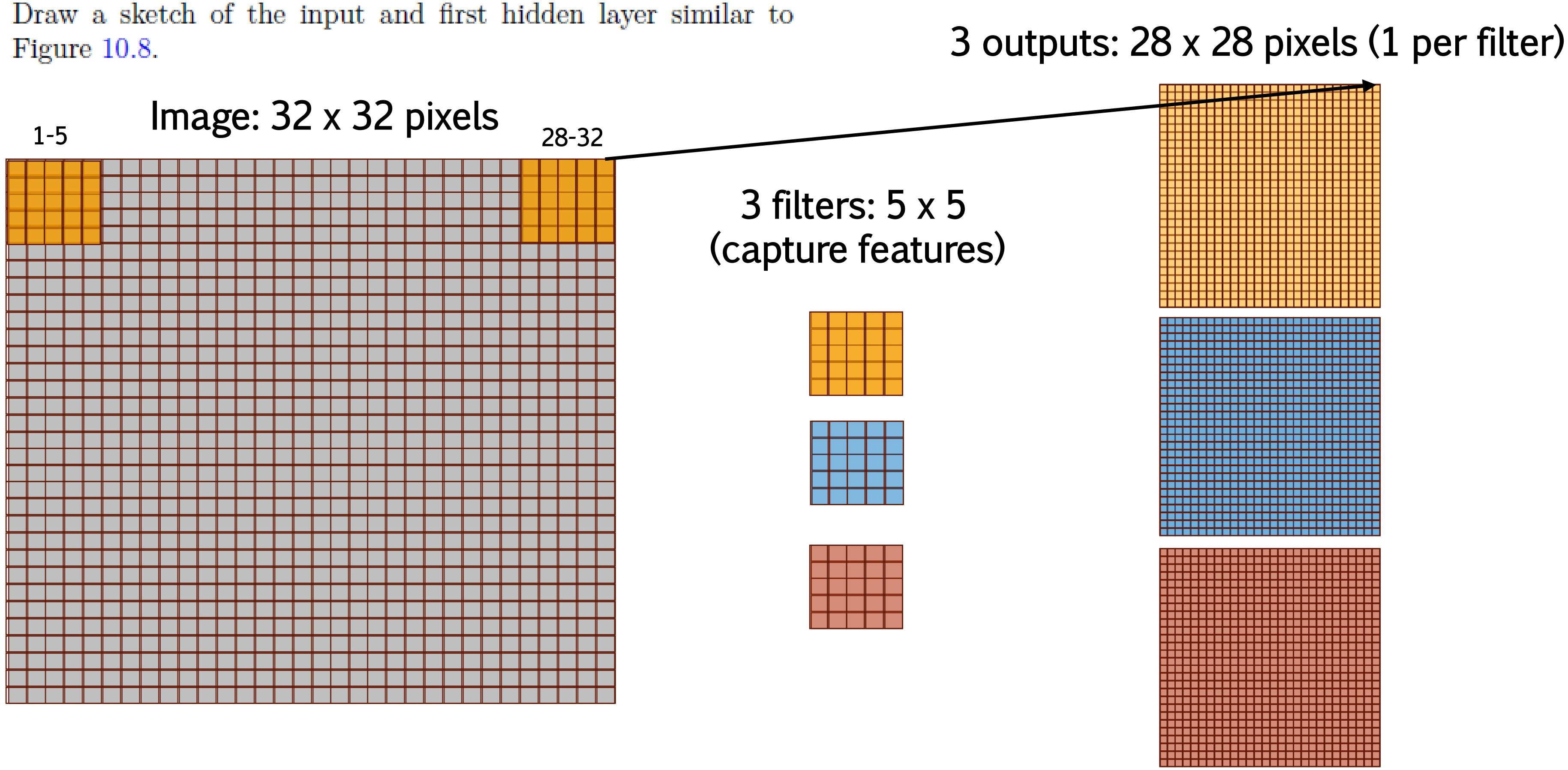


10 + 9 + 4 = 23

Ex.4) CNN: Picture (a)

4. Consider a CNN that takes in 32×32 grayscale images and has a single convolution layer with three 5×5 convolution filters (without boundary padding).

   (a) Draw a sketch of the input and first hidden layer similar to Figure 10.8.

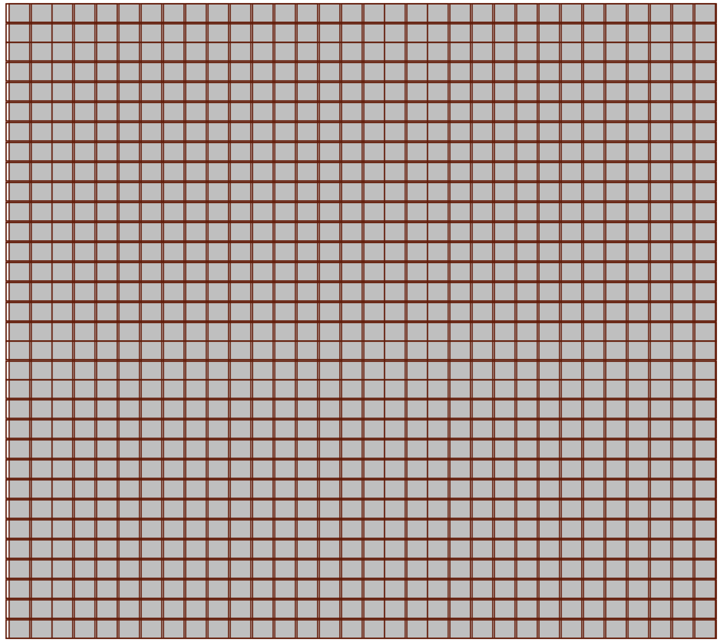


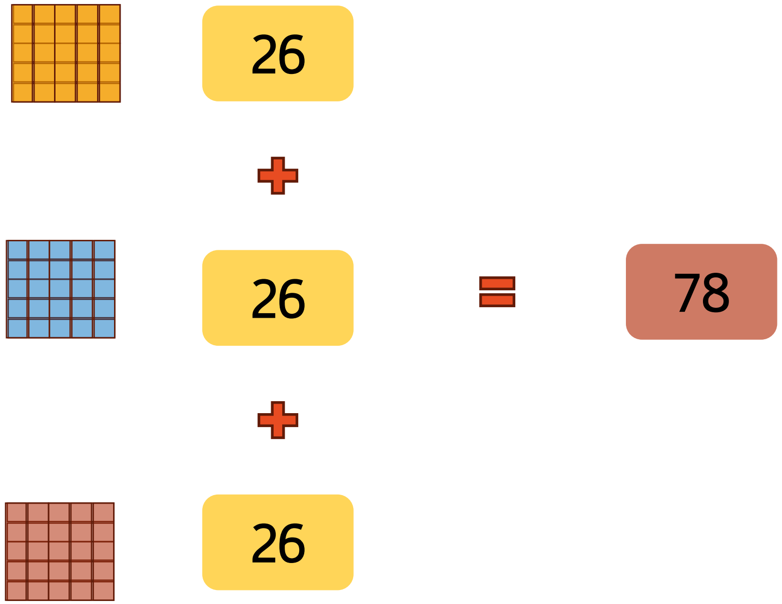
Ex.4) CNN: Parameters (b)

(b) How many parameters are in this model?

Each filter: 25 weights + 1 intercept (linear model)

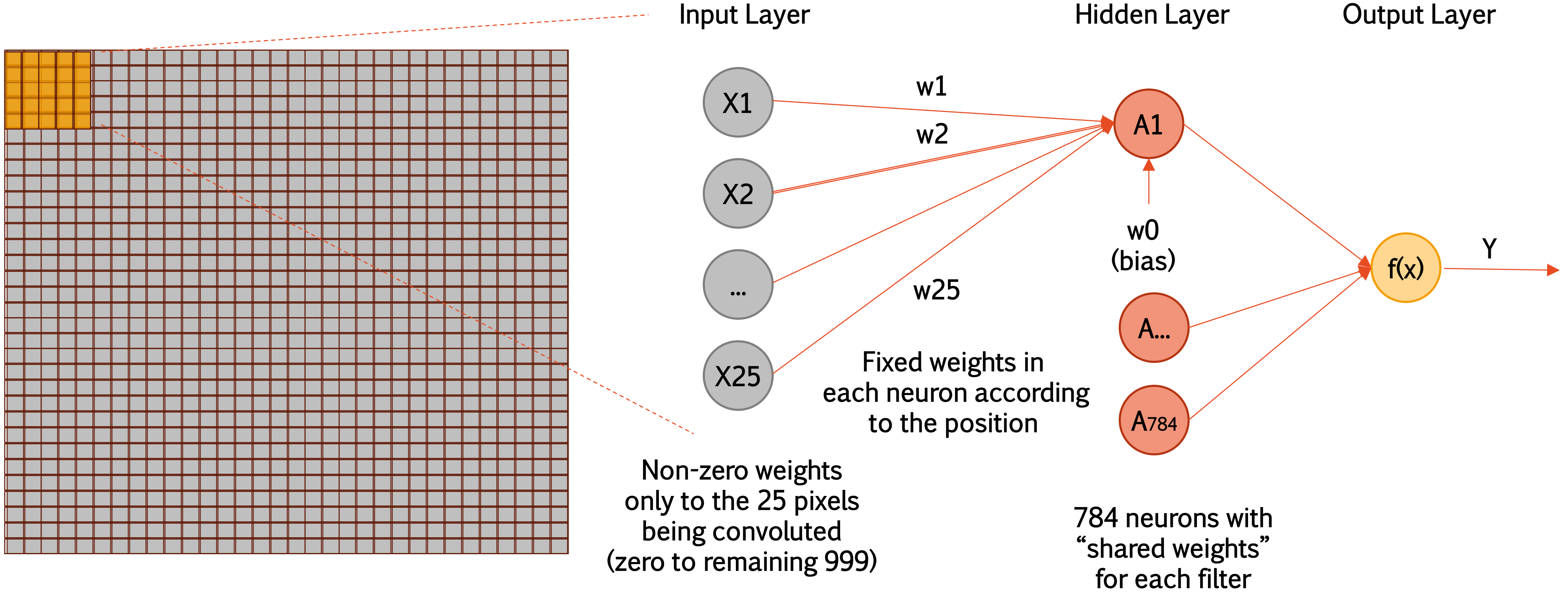
Image: grayscale → 1 channel





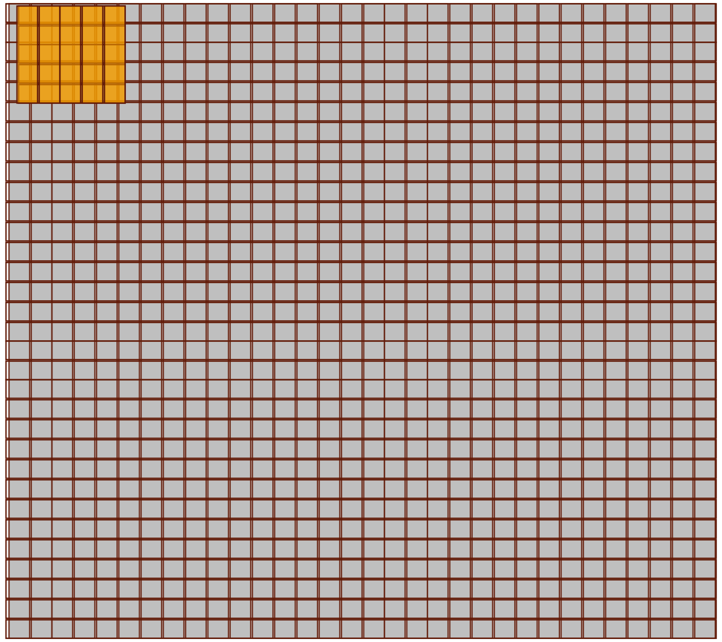
Ex.4) CNN: Constrained Feed-Forward NN (c)

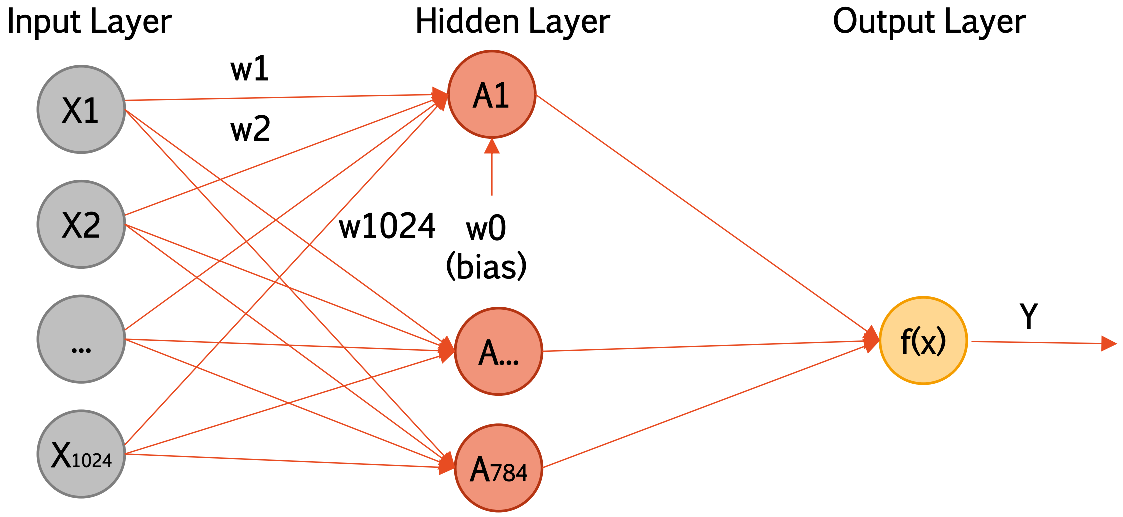
(c) Explain how this model can be thought of as an ordinary feed-forward neural network with the individual pixels as inputs, and with constraints on the weights in the hidden units. What are the constraints?



Ex.4) CNN: Constrained Feed-Forward NN (c)

(d) If there were no constraints, then how many weights would there be in the ordinary feed-forward neural network in (c)?





1024 ✖ 2352 = 2.4 millions

weights for each neuron

Neurons (784 for each filter)

weights

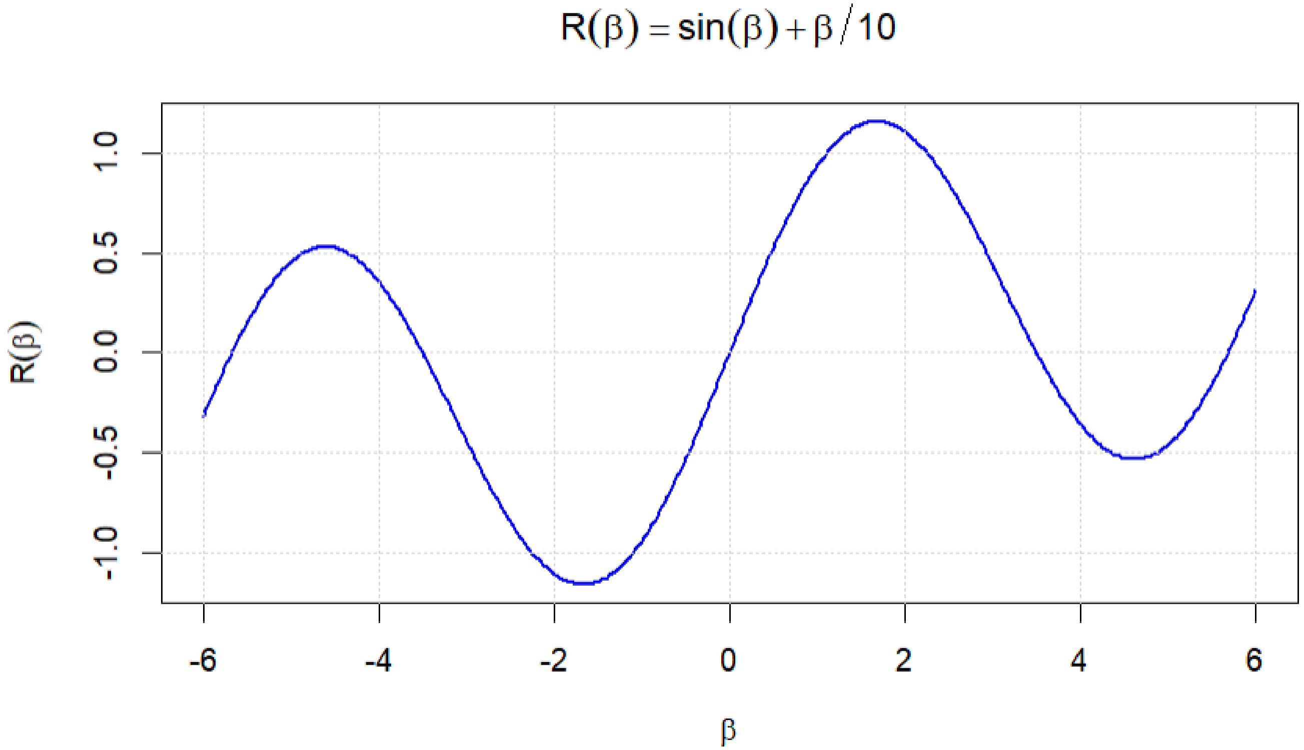
Applied

Exercise 6 – Gradient descent

Ex.6) Gradient Descent: Plot (a)

6. Consider the simple function $R(\beta) = \sin(\beta) + \beta/10$.

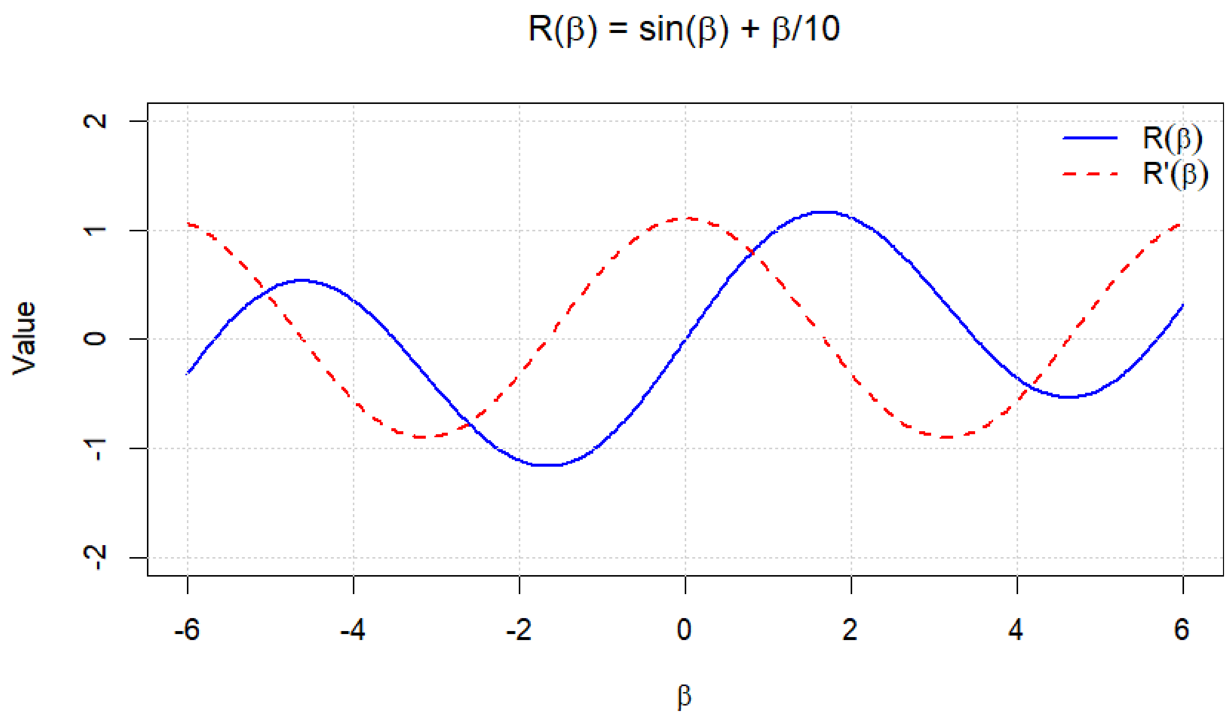
   (a) Draw a graph of this function over the range $\beta \in [-6, 6]$.



Ex.6) Gradient Descent: Derivative (b)

(b) What is the derivative of this function?

$$\frac{dR(\beta)}{d\beta} = \frac{d(\sin\beta + \beta/10)}{d\beta} = \cos\beta + \frac{1}{10}$$

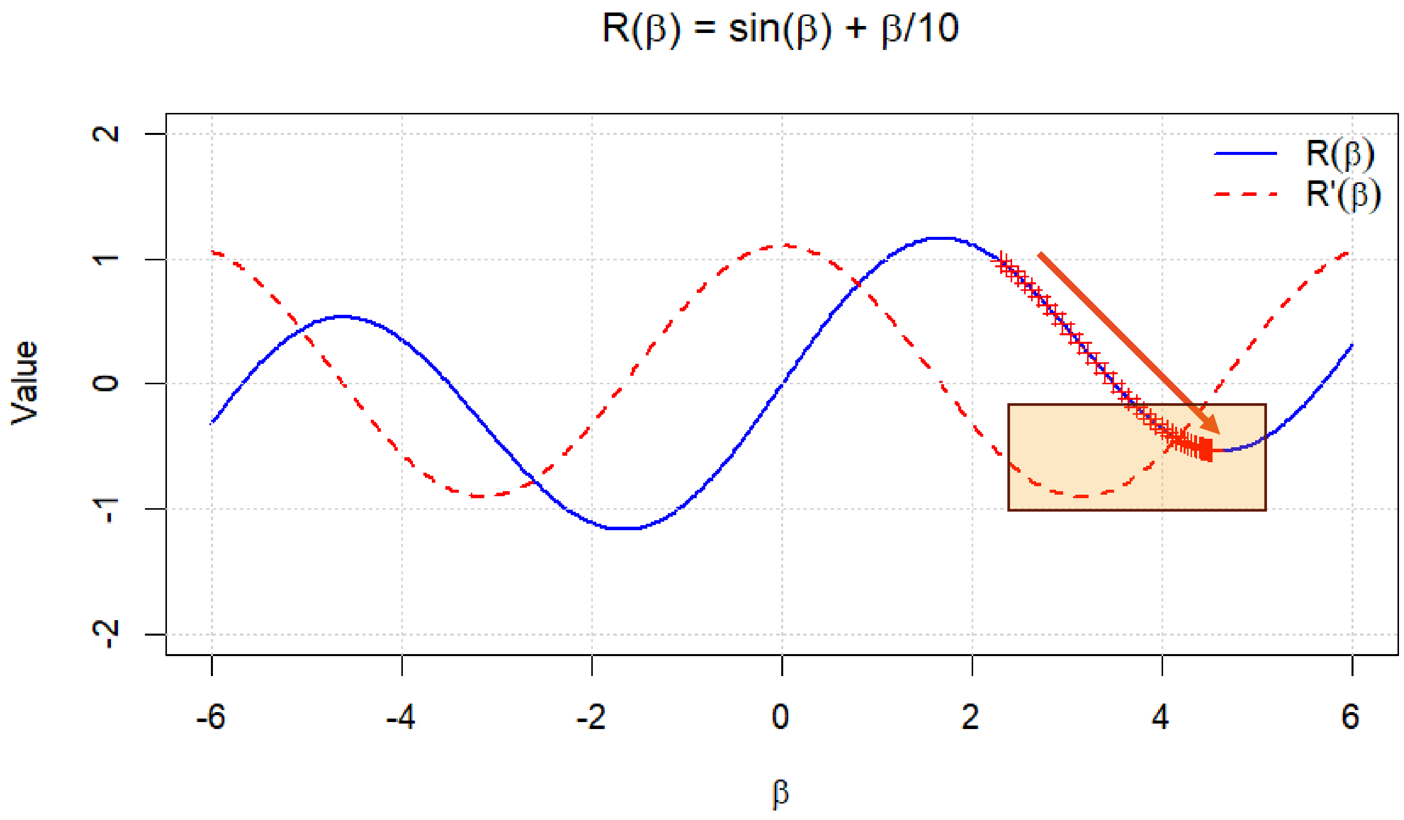


Ex.6) Gradient Descent: Negative Gradient (c)

(c) Given $\beta^0 = 2.3$, run gradient descent to find a local minimum of $R(\beta)$ using a learning rate of $\rho = 0.1$. Show each of $\beta^0, \beta^1, \ldots$ in your plot, as well as the final answer.

$$\beta^i = \beta^0 - \boldsymbol{\rho} \frac{dR(\beta)}{d\beta}$$

Find a minimum: move in the opposite direction of the gradient (**minus sign**)

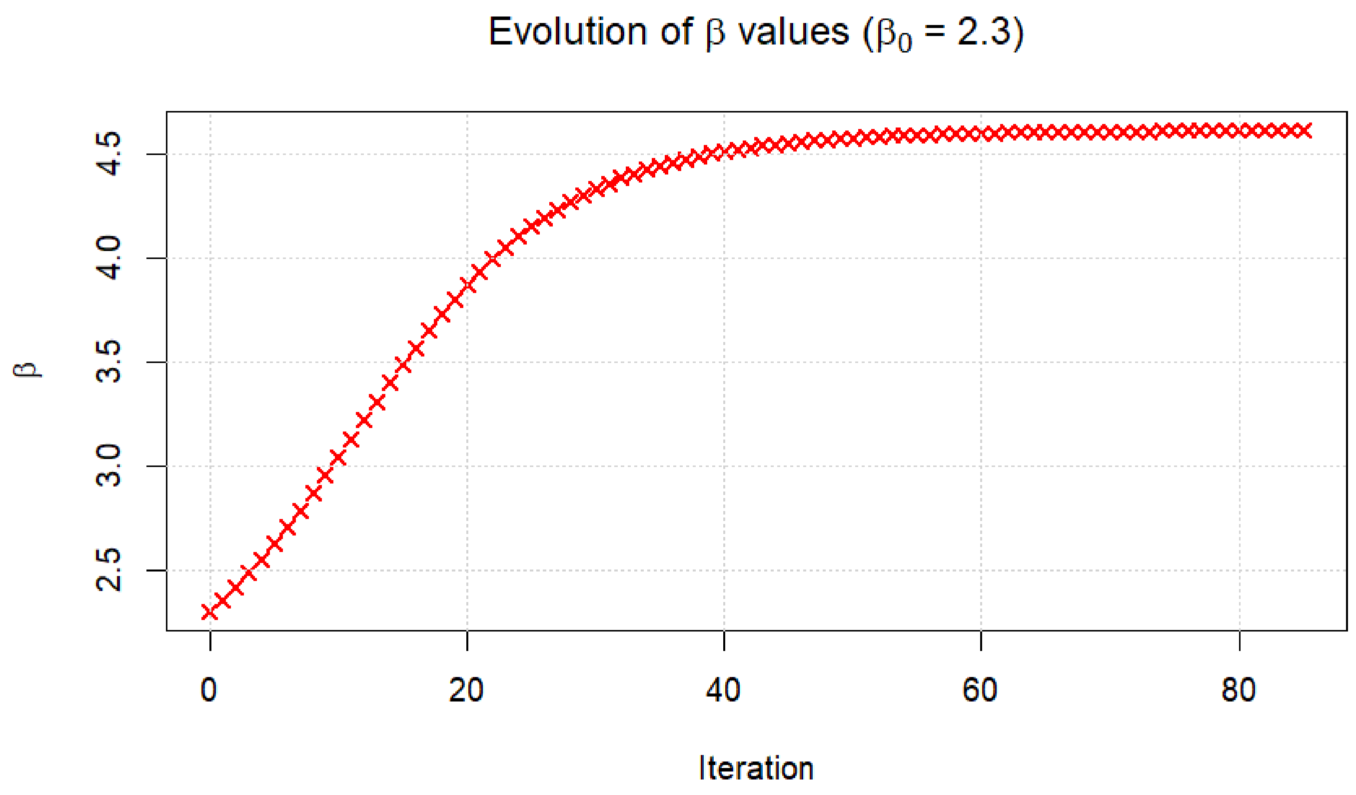


Since Gradient <0
→ Increasing Beta

Final Beta = 4.611

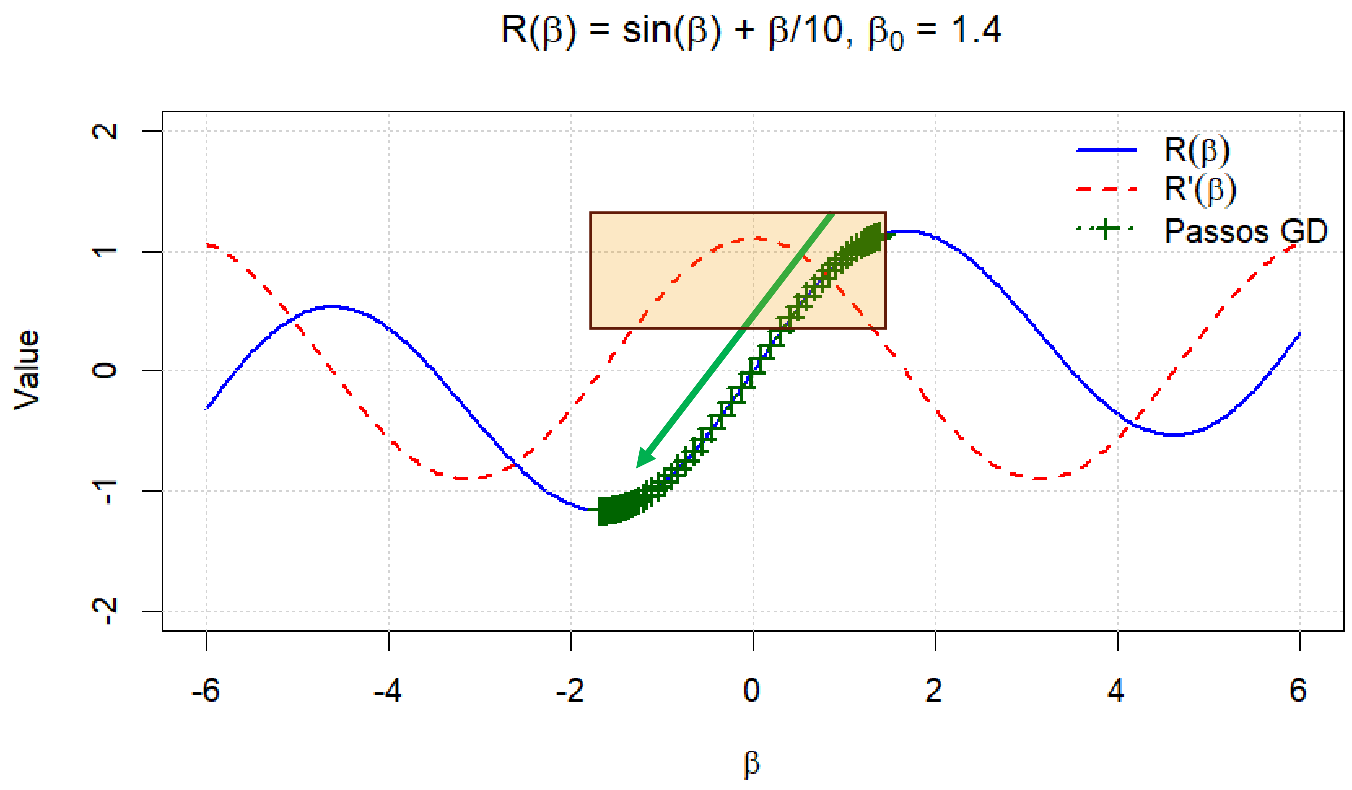
Ex.6) Gradient Descent: Negative Gradient (c)

(c) Given $\beta^0 = 2.3$, run gradient descent to find a local minimum of $R(\beta)$ using a learning rate of $\rho = 0.1$. Show each of $\beta^0, \beta^1, \ldots$ in your plot, as well as the final answer.



Ex.6) Gradient Descent: Positive Gradient (d)

(d) Repeat with $\beta^0 = 1.4$.



Since Gradient >0
→ Decreasing Beta

Final Beta = -1.670

Ex.6) Gradient Descent: Positive Gradient (d)

(d) Repeat with $\beta^0 = 1.4$.